

Mining Attention Dynamics

Auditing KV-Saliency Prediction in LLMs

**Evaluate the prediction under
the decision that will use it.**

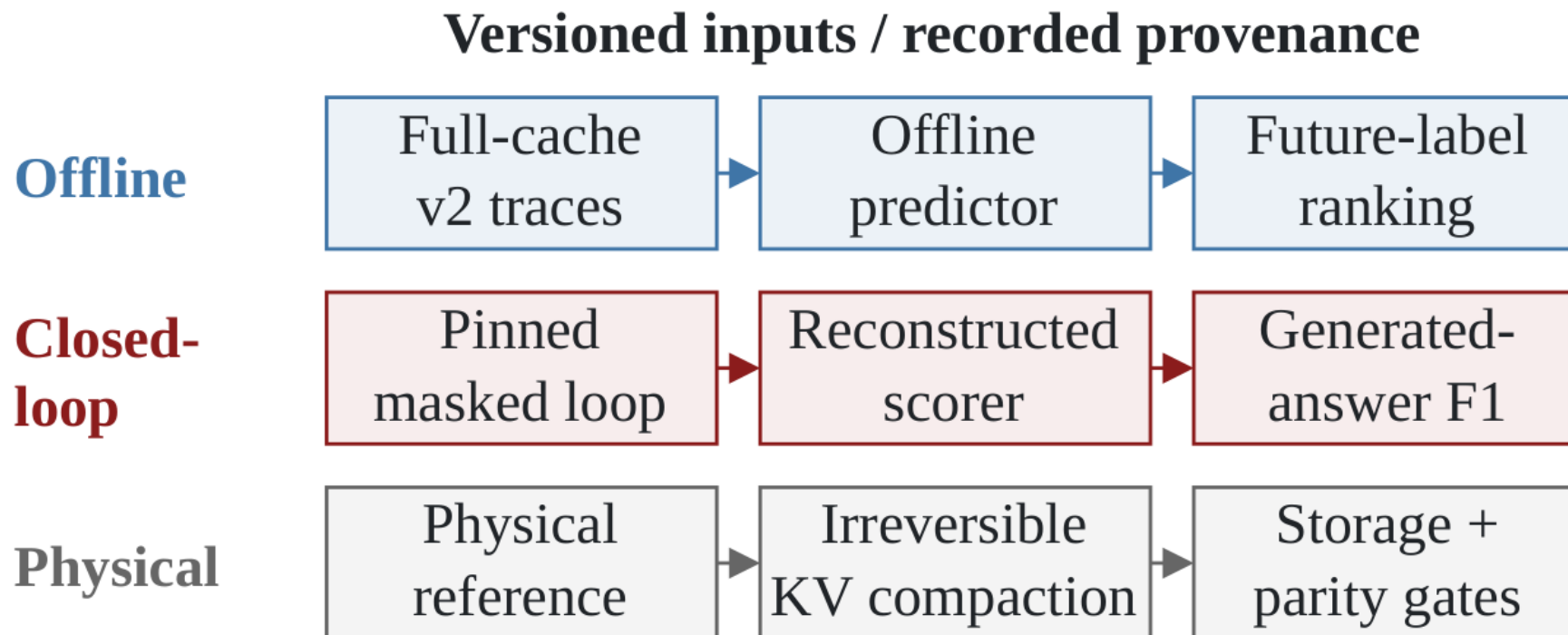
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Three paths, three evaluation objects

Feature realization and selector semantics are explicit in the protocol.



Logical retention and physical KV storage require separate measurements.

Versioned data supports the offline comparison

Main offline corpus: Llama-3.1-8B-Instruct and Qwen2.5-7B-Instruct

Protocol component	Version 2
Labeled rows	30,445,332
Source prompts / model-requests	128 / 256
Training / held-out sampled rows	120,000 / 150,000
Additional split control	Source-prompt disjoint
Collector corrections	RoPE alignment; real lookahead

Request and source-prompt units matter more than the row count.

Source: icdm_v2.json · paper §V-A

Pooled AUC favors the learned predictors

Two features and one intercept give a compact offline scorer.

Offline score	Pooled AUC	Pooled AP
Within-layer EMA	0.873	0.609
Two-view logistic	0.890	0.667
LightGBM, balanced	0.896	0.674

A pooled ranking does not enforce a separate budget at every decision.

Source: icdm_v2.json / pooled_request_split.table · paper Table II

The ordering changes under a decision budget

Pooled AUC is almost entirely cross-decision pairs; a selector uses the order inside each decision

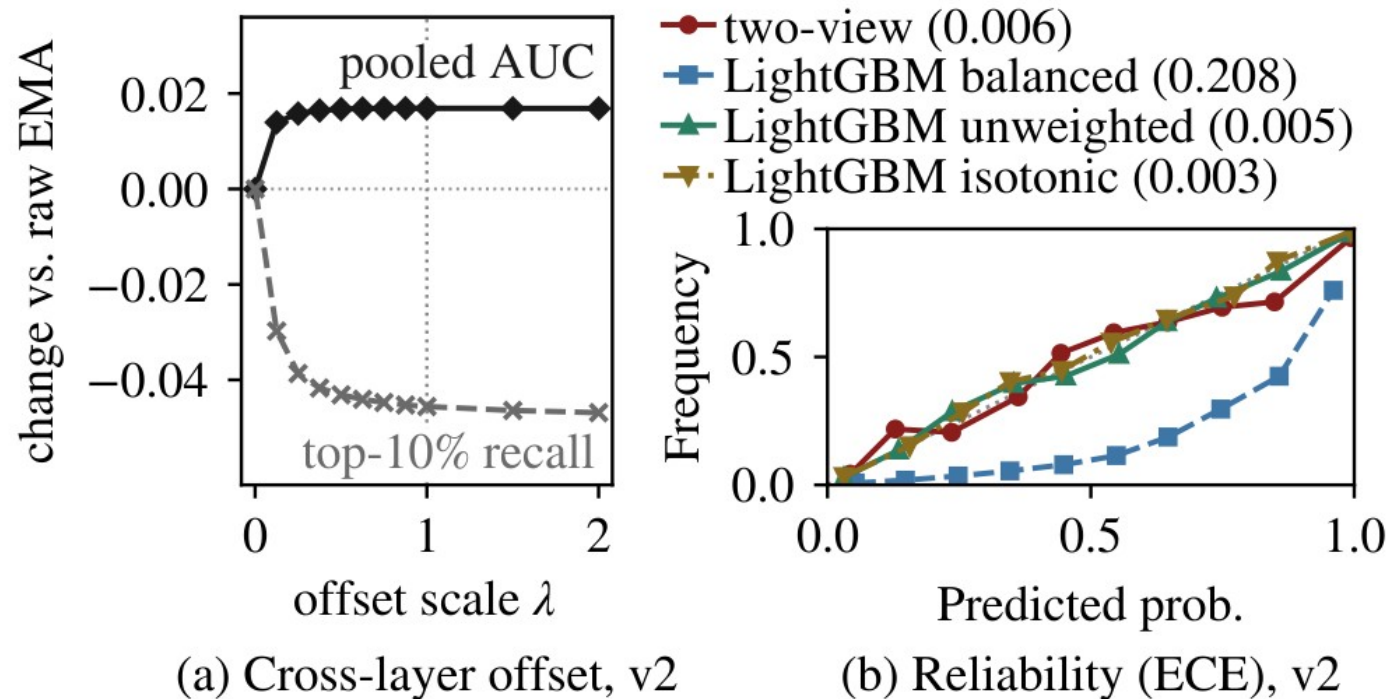
Offline score	Pooled AUC	Within-decision AUC	Top-decile recall
Within-layer EMA	0.873	0.930	0.659
Two-view logistic	0.890	0.928	0.613
LightGBM, balanced	0.895	0.925	0.615

A feature can earn its place pooled and cost recall inside the decision.

Source: icdm_v2_decomposition.json (every held-out row) · paper Table II, lower block

Offset sweep and calibration

(a) Scaling the cross-layer offset trades pooled AUC for recall; (b) calibration depends on the fit.



Low ECE does not establish a superior retention policy.

The runtime scorer reconstructs its features

A matched-row bridge separates feature realization from trajectory changes.

Feature / trajectory arm	Llama bf16	Qwen fp32
A: offline / full cache	0.902	0.894
B: runtime / full cache	0.899	0.892
C: runtime / on policy	0.989	0.989

The bridge measures reconstruction; the historical gap's cause remains open.

Source: feature_bridge/bridge.{llama.bf16,qwen.fp32}.json · paper §IV-B

Closed-loop answer quality: the pinned rerun

448 source prompts × 2 models = 896 model–prompt evaluations; 7 datasets

20% logical retention; each policy drives its own trajectory.

Pinned simulator source and bfloat16 numerics.

Official LongBench scoring with all reference answers.

Primary: TOST over seven dataset means; practical ± 0.02 F1 margin, not pre-registered.

The main task-quality comparison has two architectures and seven tasks.

Source: `expand_v2/` provenance and `tost/expand_v2_sensitivity.json` · paper Exp#8

Mean F1 is equivalent within the chosen margin

Both compressed policies remain below full-cache quality.

Policy realization	Mean F1
Full cache	0.419
H2O-style accumulation	0.372
Reconstructed two-view	0.374

Paired difference +0.0017 | seven-dataset TOST $p = 0.0023$

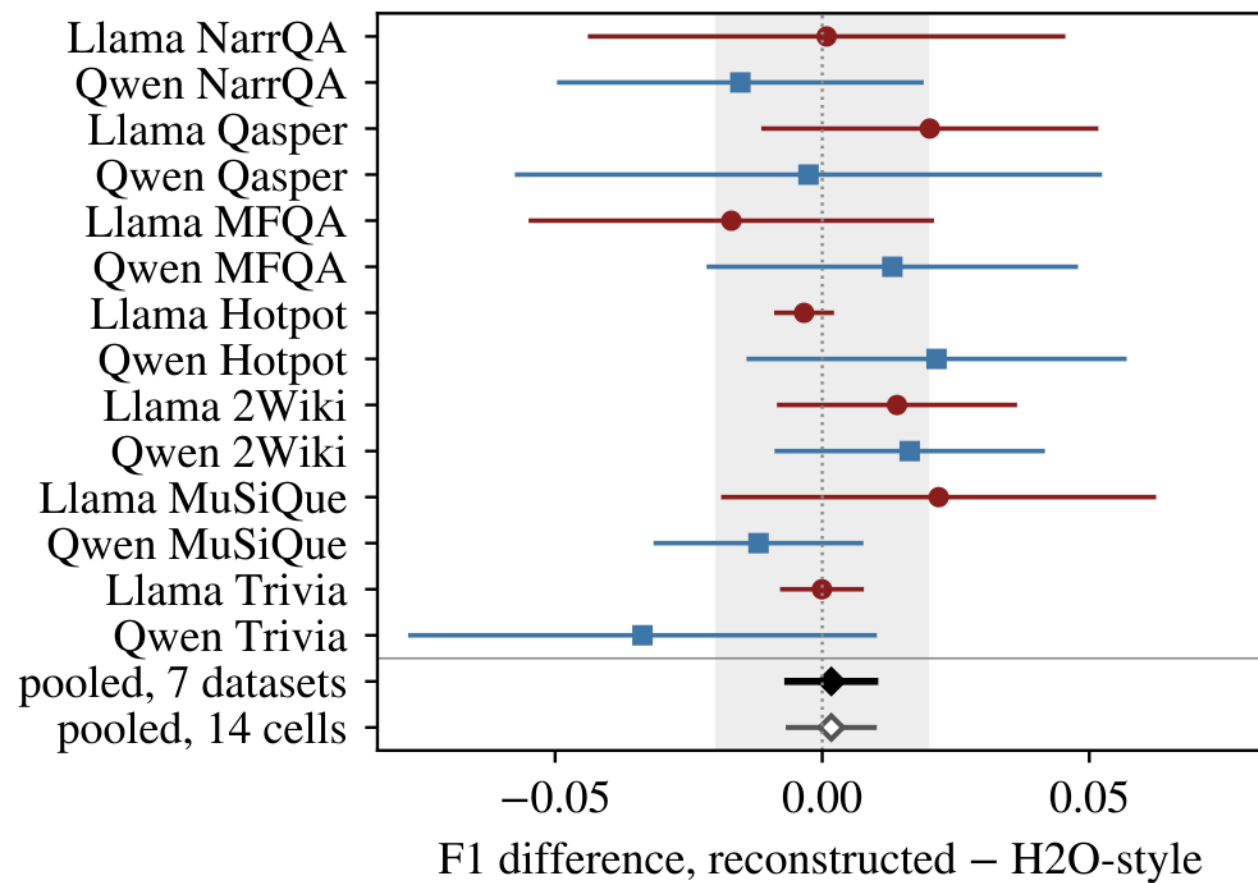
Seven-dataset 90% t interval [-0.0064, +0.0098]

14-cell sensitivity: $p = 0.0007$, 90% t interval [-0.0063, +0.0097]

Source: `tost/expand_v2_sensitivity.json` · paper Table IV

Task heterogeneity remains visible

An average-equivalence result can contain gains and losses by task.



Source: `tost/expand_v2_sensitivity.json` · Python-generated paper forest plot

Physical storage falls; total memory falls much less

A separate eager, single-request reference backend at 20% block budget

Model	Stored KV (MiB)	Decode peak (GiB)
Llama bf16	496 → 94	15.51 → 15.10
Qwen fp32	434 → 82	29.05 → 28.63

Extended mask/physical parity exceeds the preset numerical tolerance.

No production TPOT, concurrency or throughput conclusion follows.

Source: `physical_kv/{llama3.1-v1,qwen25-v1}/summary.json` · paper Table III

Query evidence is scoped to the tested controls

Phase-aligned dot-max is a probe; it is not Quest's page-bound algorithm.

Model	Request-mean AUC change
Llama-3.1-8B	-0.0070
Qwen2.5-7B	+0.0020

No extrapolation to MoE, base models, $\geq 32K$ contexts or sampling.

Source: physical_kv*/summary.json / query_v2 · paper §IX

A benchmark contribution to data systems for AI

Versioned collection, provenance and explicit evaluation contracts

1. Evaluate scores under the policy's decision budget.
2. Specify feature reconstruction and selector semantics.
3. Separate answer quality, physical storage and serving cost.
4. Preserve provenance and report the limits of each experiment.

The default protocol scores pooled and per-decision metrics together.

Code and paper: github.com/lzbaclz/KVSaliencyBench · corresponding author: chenjx@hust.edu.cn